PurityLoop AI

Smart waste classification for cleaner recovery decisions

CAPSTONE PROJECT 2026

AI-powered waste classification with human review and analytics

SUNWAY UNIVERSITY | Sunway BUSINESS SCHOOL

Mixed waste. Verified recovery.

Browser detection, backend video processing, review control, and analytics in one operational workspace.

Problem + Objective

- Uncertain labels
- Mixed waste
- Verified records

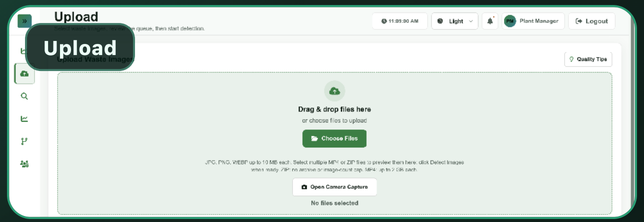
Solution

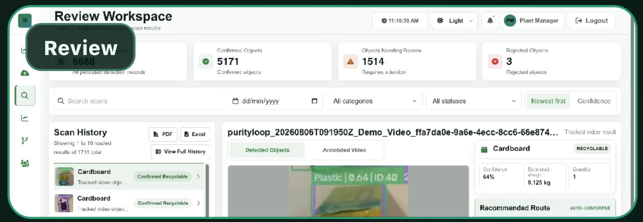
AI Detect > OK Review > DB Record

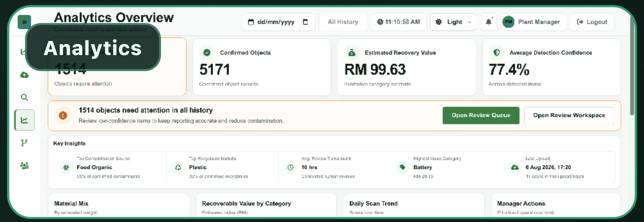
Limitations

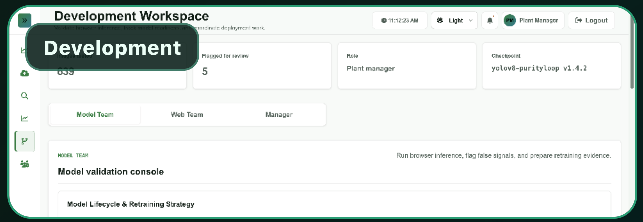
- L — Lighting variance
- O — Occluded waste
- M — Mixed materials
- D — Dataset drift
- R — Review workload

Web Interface + Dashboard Evidence









Project Lifecycle + Tool Integration

AI model	Business understanding	Data preparation	Modelling	Evaluation	Deployment
Web app	Requirements	System design	Prototype	User review	Refinement
Analytics	Scan records	Material rows	KPIs	Insights	Actions

Next.js	TypeScript	FastAPI
Python	Supabase	Vercel
Cloud Run	Cloud Tasks	Google Drive
ONNX Runtime	YOLOv8m-seg	Browser AI

Redrawn from the provided lifecycle so it fits the dark poster system.

Key Metrics / Results

- **77.2%** Overall mAP, 9-class mean box mAP@0.5
- **81.3%** Peak baseline, single-model benchmark
- **90.6%** Naming accuracy, detected-item label check
- **182k+** Generalisation image corpus

Target Audience

- FM — Facility managers
- CS — Campus teams
- WO — Waste operators
- PR — Project reviewers

Group Members

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Model Framework

YOLOv8m-seg
ONNX

- UP — Upload
- AI — Detect
- OK — Review
- KP — Analytics
- AD — Admin

Future Actions

More data coverage. Better retraining. Robotic sorting integration.